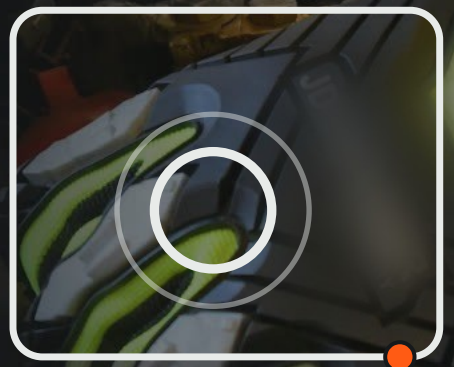
praxis

STEP 4/9

Instructions that can finally see how you're doing

Software for smart glasses that coaches new maintenance technicians from a company's existing manuals.



Seat the clamp past the hose bead

Startups @ Harvard · October 20, 2026 · Monish Saravana

Problem

Training new technicians is expensive

Each new hire learns by shadowing an experienced technician, and that time comes off the job.

Total cost to hire and ramp up one maintenance technician

$10–15k

including recruiting fees, onboarding time and lost productivity

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~255,000

US maintenance and repair openings a year, each one needing someone to train them

In a 2017 survey of 35 aviation repair stations, new hires took 9–14 months to become profitable.

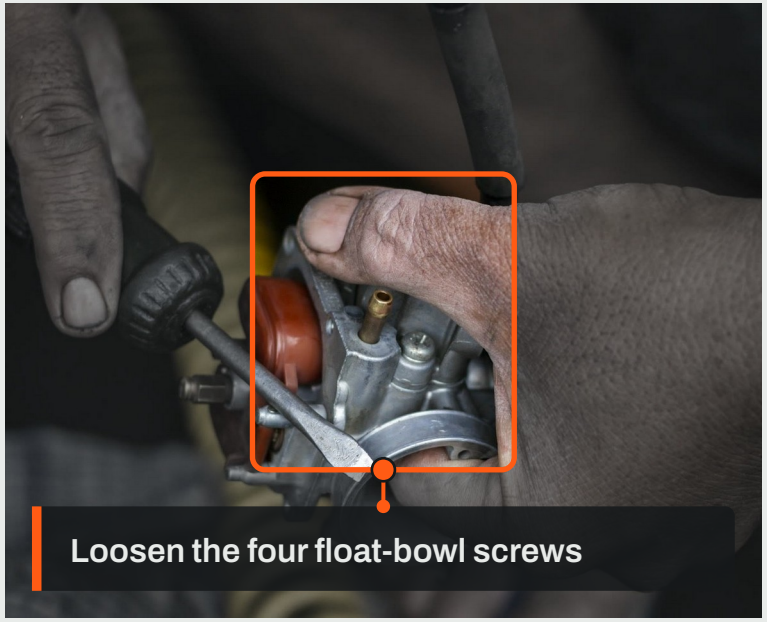
> **“The hardest AND most expensive part is training new techs …”**
> Casey Czarnowski, Healthcare Sterile Processing Association (HSPA), discussed this week



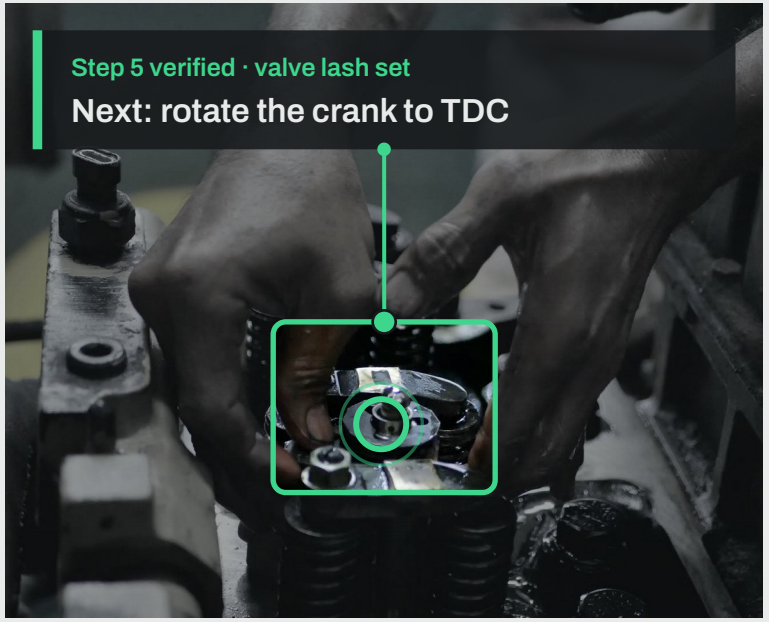
Praxis coaches the job through smart glasses

1 Upload the manual → 2 Praxis compiles it → 3 Tech wears glasses → 4 Praxis tracks the job → 5 Verifies key steps

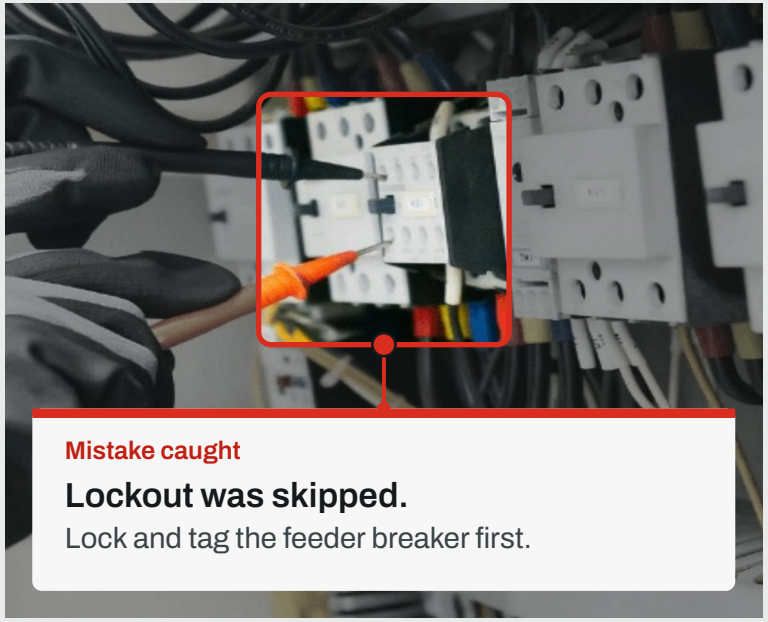
1 Says the next step



2 Checks the work



3 Catches a mistake



Built for new technicians in plant maintenance, facilities, HVAC service and equipment repair.

This is already a working software system

1 A manager uploads the manuals they already have

3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
FIST 1-1 · Hazardous Energy Control Program FIST_1-1.pdf · v1	97	131	4
FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

2 Praxis compiles a procedure; the manager reviews and approves it

High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live ▾ · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT... ×
- **high voltage** Observe the high voltage board for a moment ×
- **high voltage** Take the high voltage board and places it on the work area ×
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts ×
- **disconnect the power** Disconnect the power supply cables ×
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th... ×
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1... ×
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160... ×
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts ×

STEPS 20 · 0 safety-critical — Expand all

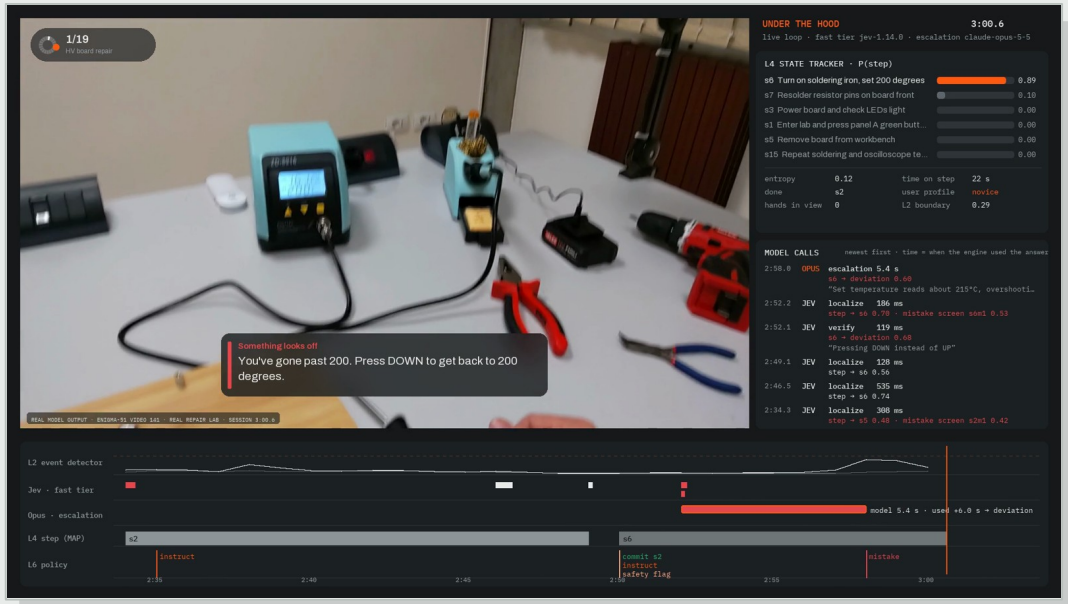
1 **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card ›

What the compiler saw — 6,569 chars

Page 1 ↗

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back

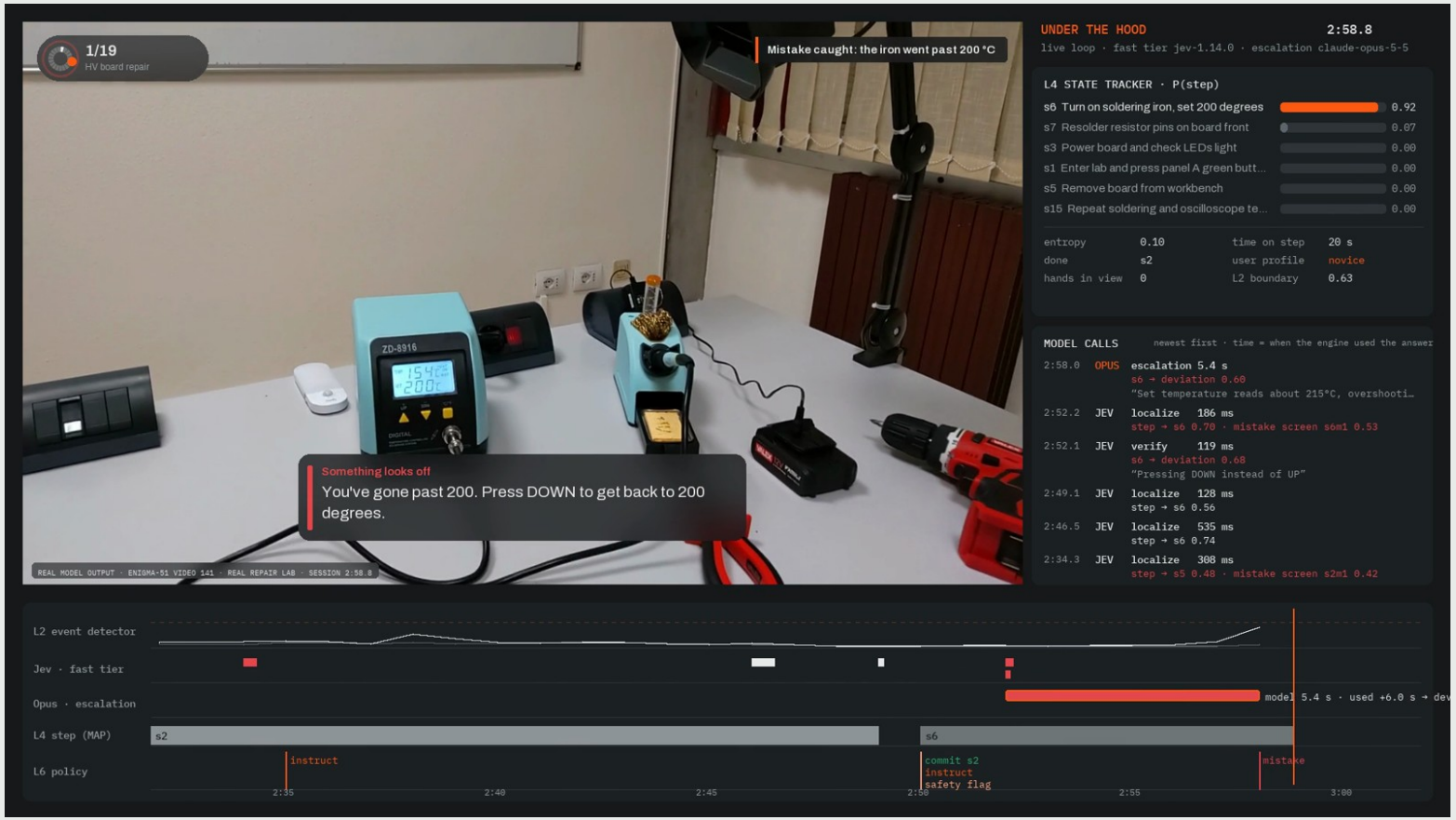
3 The live app tracks the job (engineer view)



- **Company backend: upload, index, compile, approve** built
- **Tracking engine and evaluation harness** built
- **Live app with spoken coaching** built, real time on video
- **Meta glasses adapter** coded, not device-tested
- **Viture/NVIDIA glasses adapter** coded, not device-tested
- **Android XR glasses** planned

Live test

We already run a real repair procedure live



Real repair, start to finish

16:40

a high-voltage board repair, filmed first-person in an industrial lab

Training on this exact job

None

Praxis started from the lab's written procedure

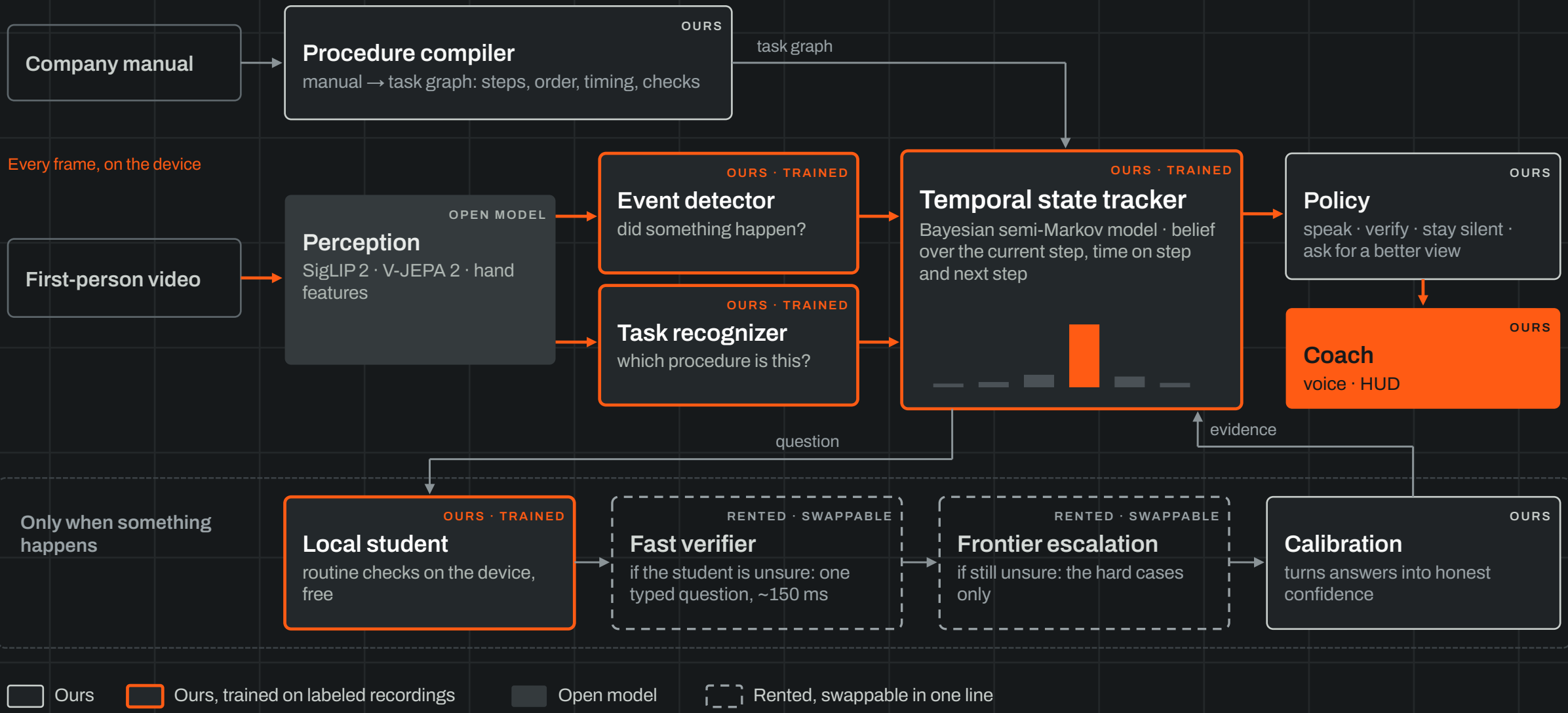
Model cost

≈$1

of model calls per coached hour, measured across our evaluation

How it works

The system tracks the job instead of querying a model constantly

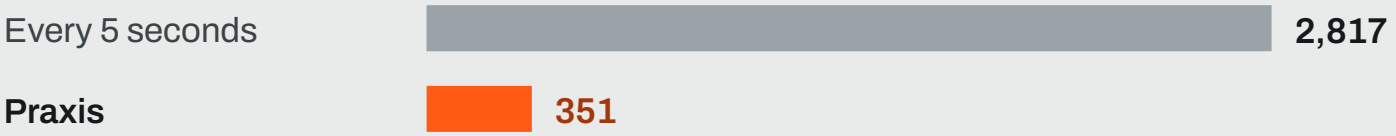


Measured

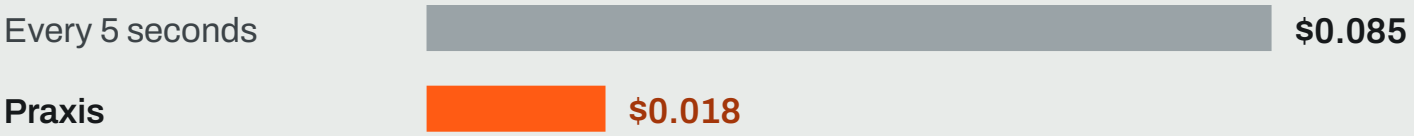
Same frontier model, better system

We compared Praxis with calling the same model (Claude Opus 5.5) every 5 seconds, on 48 first-person videos neither had seen.

8× fewer calls to the frontier model



5× cheaper per minute of video



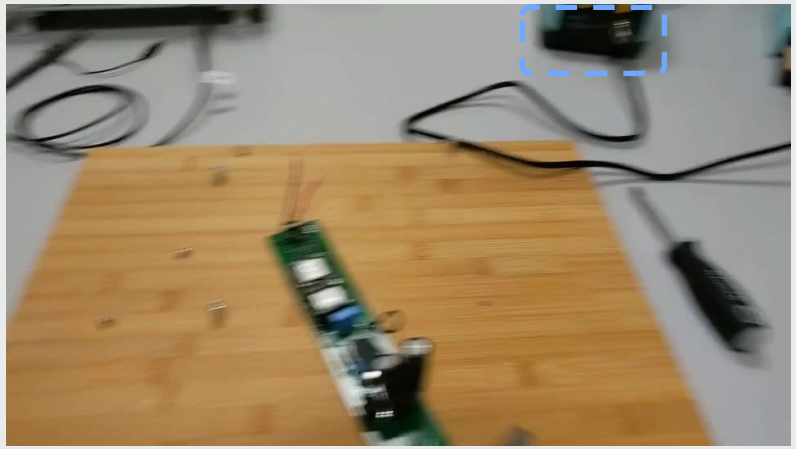
About $1

of model calls per coached hour, against $200 a month per trainee

Better tracking

It follows the step more accurately than the baseline while asking the frontier model 8× less

When it can't see, it says so



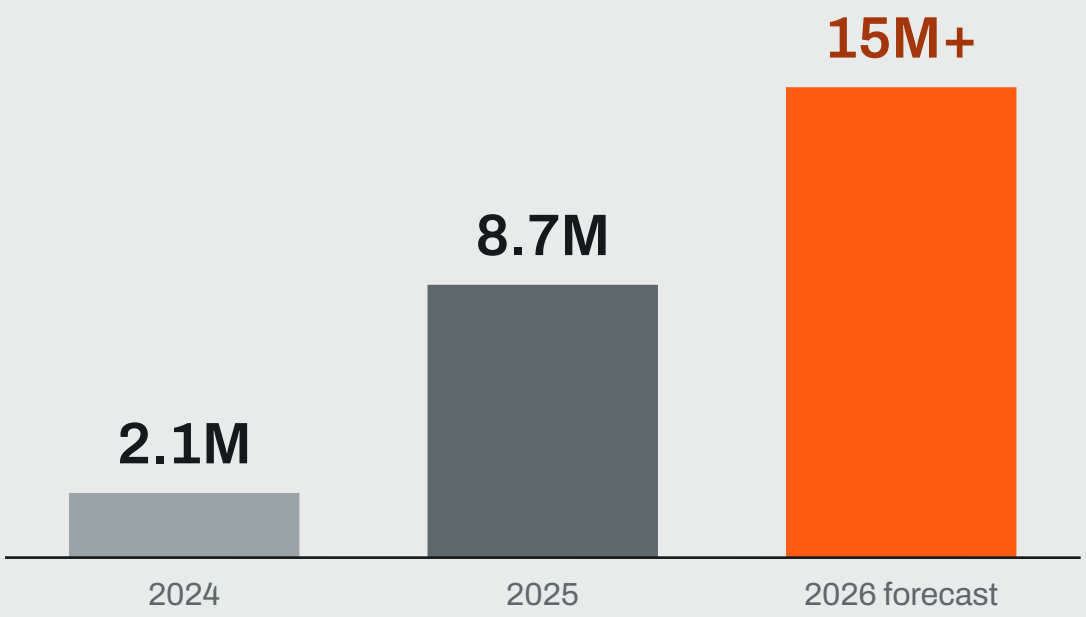
> **“Look closely at the soldering station display, upper right.”**
>
> Praxis at 4:25 in the live repair, with the display half out of frame

77–98% of blurred or blocked checks: Praxis asked to see it again

Up to 73% the every-5-seconds baseline named the wrong step with confidence

Smart-glasses hardware is finally open to software

AI glasses shipped per year, worldwide

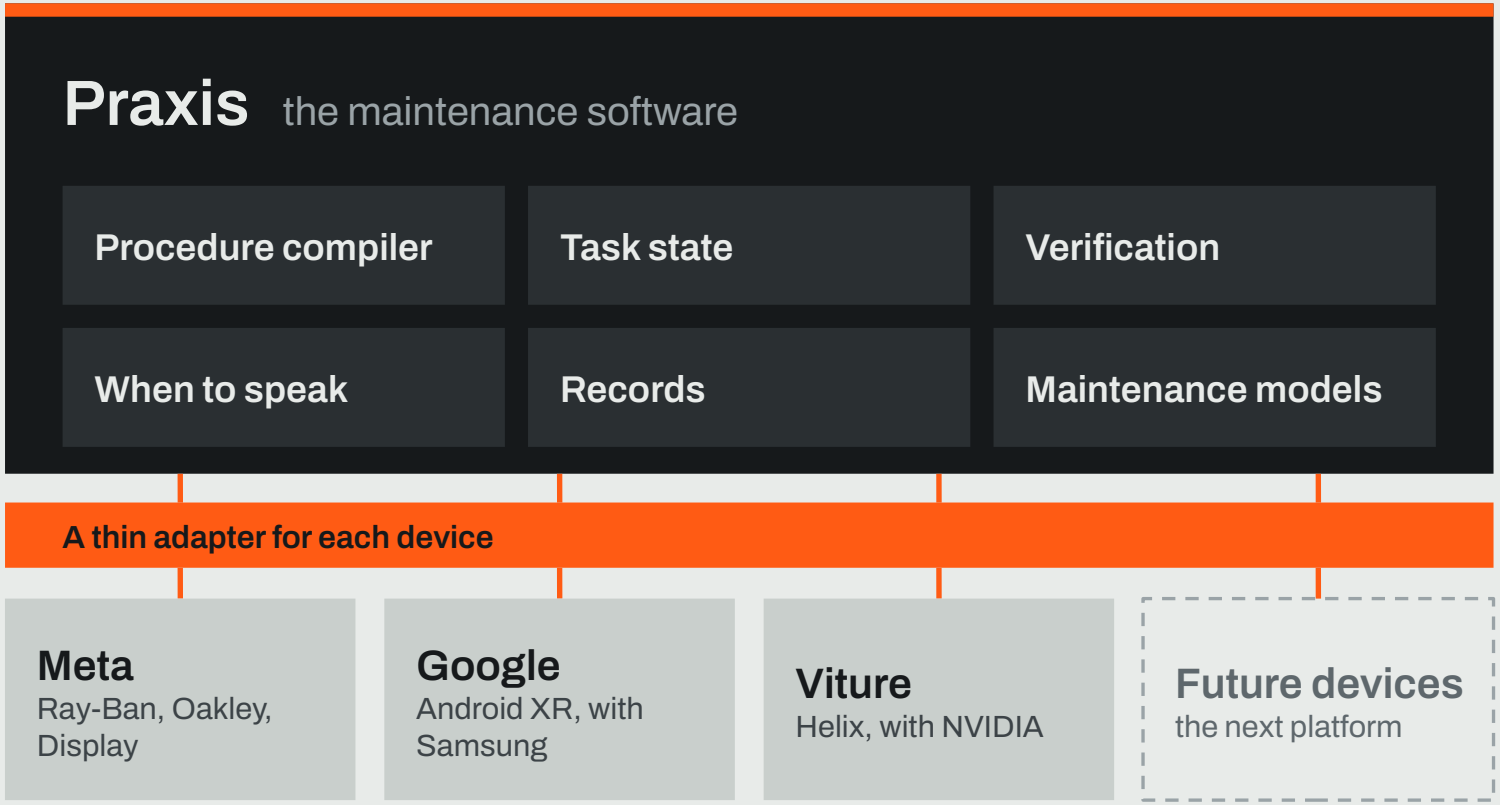


Platforms opening to outside developers

2025	Meta opens a developer preview: outside apps can use its glasses' camera
Sept 2026	Meta's toolkit reaches 1.0: camera, microphones and speakers for outside apps
Fall 2026	Google's first Android XR glasses arrive: audio, with a camera
Q1 2027	Viture and NVIDIA ship Helix: safety glasses that coach workers on procedures in real time

Three years ago we would have had to build the glasses.
Now we build the software that runs on them.

We don't need to win the glasses hardware race



They build the glasses. Every new device is one more adapter for Praxis.

Other approaches

Procedure software MaintainX, Tulip, Augmentir
Steps on a screen. It can't see the work.

Fixed cameras Retrocausal, Invisible AI
Watch one station. They can't follow a tech around a machine.

General assistants Meta AI, Gemini Live
Answer about what's in view, with no procedure or step state.

Viture Helix + NVIDIA Our closest competitor
Real-time coaching sold with its own glasses. We want Praxis to run on Helix too.

Maintenance is where we start, and Praxis pays for itself

Maintenance new-hire onboarding in the US, per year

~255k new maintenance techs × **$200** a month per trainee × **~6** months of onboarding ≈ **$306M** a year, at our pricing hypothesis

- **Start** maintenance new-hire onboarding
- **Next** all maintenance and field service
- **Then** technical onboarding in manufacturing, utilities, aviation, telecom, labs

One new technician

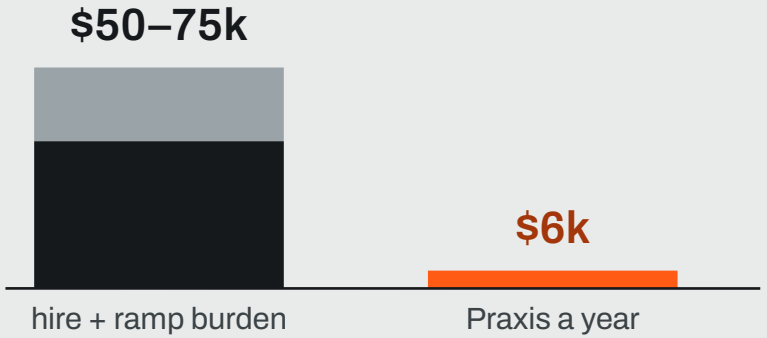
Hire + ramp burden today	**$10–15k**
Praxis, six months	**$1,200**
Our cost to serve	**~$270**

~77% gross margin today, ~90% with our local model

Pays for itself if it removes roughly 8–12% of that burden.

A 50-technician maintenance team

about 5 new hires a year



- **We start with small and mid-size maintenance teams.** 98% of US manufacturers have fewer than 500 employees, and there one maintenance manager can approve a paid pilot. The pilot measures senior-tech hours, days until a new tech works alone, and rework.

We sell to maintenance managers, one paid pilot at a time

1 **Maintenance manager**

at a plant, facilities team, HVAC company or repair shop

2 **Demo on their own manual**

a procedure their new techs already struggle with

3 **Short paid pilot**

the success metric is agreed first, like days until a new tech works alone

4 **Annual agreement**

$200 a month per active-trainee license

5 **Expand**

every new hire, then other sites

How pricing works

One active-trainee license = one employee being onboarded: $200 a month, unlimited coaching sessions

Five techs onboarding at once = five licenses. When one finishes, the license moves to the next hire

Glasses are shared site equipment. 10 licenses with 3 people coached at the same time needs about 3 pairs

We don't resell hardware. Sites buy supported glasses directly

Second path: trade schools

Students learn hands-on skills with Praxis and bring it to their first employer.

> **“We would be open to an experiment like this.”**
>
> Sonny Hunt, COO, The Fiber School (FiberOptic.com), discussed this week

Founder

I've built real-time vision systems for the physical world, and I've helped grow startups

Engineering and research

60,000+
video frames

LipTrac
Real-time visual speech recognition. Conv3D + bidirectional GRU, tested with 12 participants. Provisional patent filed.

217.9%
better collection than conventional methods

OCTOPAS
Autonomous oil-spill cleanup: aerial computer vision, path planning and marine robotics. Published; provisional patent filed.

7 years
FTC team captain

Competitive robotics
World Championship qualifier. Autonomous control, sensor fusion, OpenCV and path planning. My open-source FTC Coach has 25,000+ users.

Startups and growth

$50k
MRR, from under $1k

Qurios Ventures · co-founder, CMO
Built a 14,000+ member community in about four months. Acquired by Crimson Education for $1.3M.

13
person growth team

Claisen Health · co-founder, CMO
Ran acquisition across UGC, influencers and Reddit as the company raised $200k from angels.

$300k
attributable revenue

Acely AI (YC W18)
One of 6 marketing interns picked from 1,500 applicants.

Harvard College Applied Mathematics + Computer Science

Coaching comes first. Our own models and skill records come next

Now

Maintenance coaching and verification

Paid pilots. New techs coached from their own company's manuals.

Next

Maintenance-specific models

With customer permission, real maintenance recordings make our state and checking models better and cheaper.

Then

Verified skills and certification partners

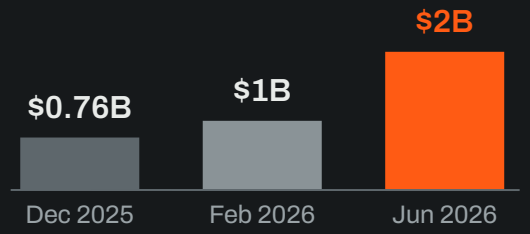
Techs and employers build verified skill histories in Praxis. Certified techs bring Praxis to their next job.

Later

Physical AI, two ways in

- **A Build our own robot models** trained on step-verified recordings customers opted in
- **B Sell to physical-AI labs** license consented data and our step checker

What AI labs pay for human training data keeps climbing



Mercor's run-rate from selling expert human data to AI labs

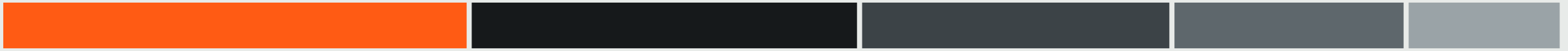
>$100M
a year that robotics companies already spend buying real-world data

>$1B
that Figure has committed to data and compute over the next 12 months

$3.2B → $16.3B
the AI training-data market, 2025 to the 2033 forecast

Ask

$10k gets us to the first paid pilot



$3,000

Model training

GPU time to retrain the step tracker, event detector and local checker on real maintenance footage

$2,500

Devices + capture

Supported glasses and capture gear to run Praxis end to end at a real site

$2,000

Maintenance data

Pay techs and trade-school students to record and label ~80 hours of real work, with consent

$1,500

Pilot inference

About 1,400 coached hours at today's measured cost

$1,000

Company + legal

Formation and the contracts a paid pilot needs

By April

Models retrained on real maintenance footage

Praxis running end to end on real glasses at a real site

A first paid pilot, with the success metric agreed before launch